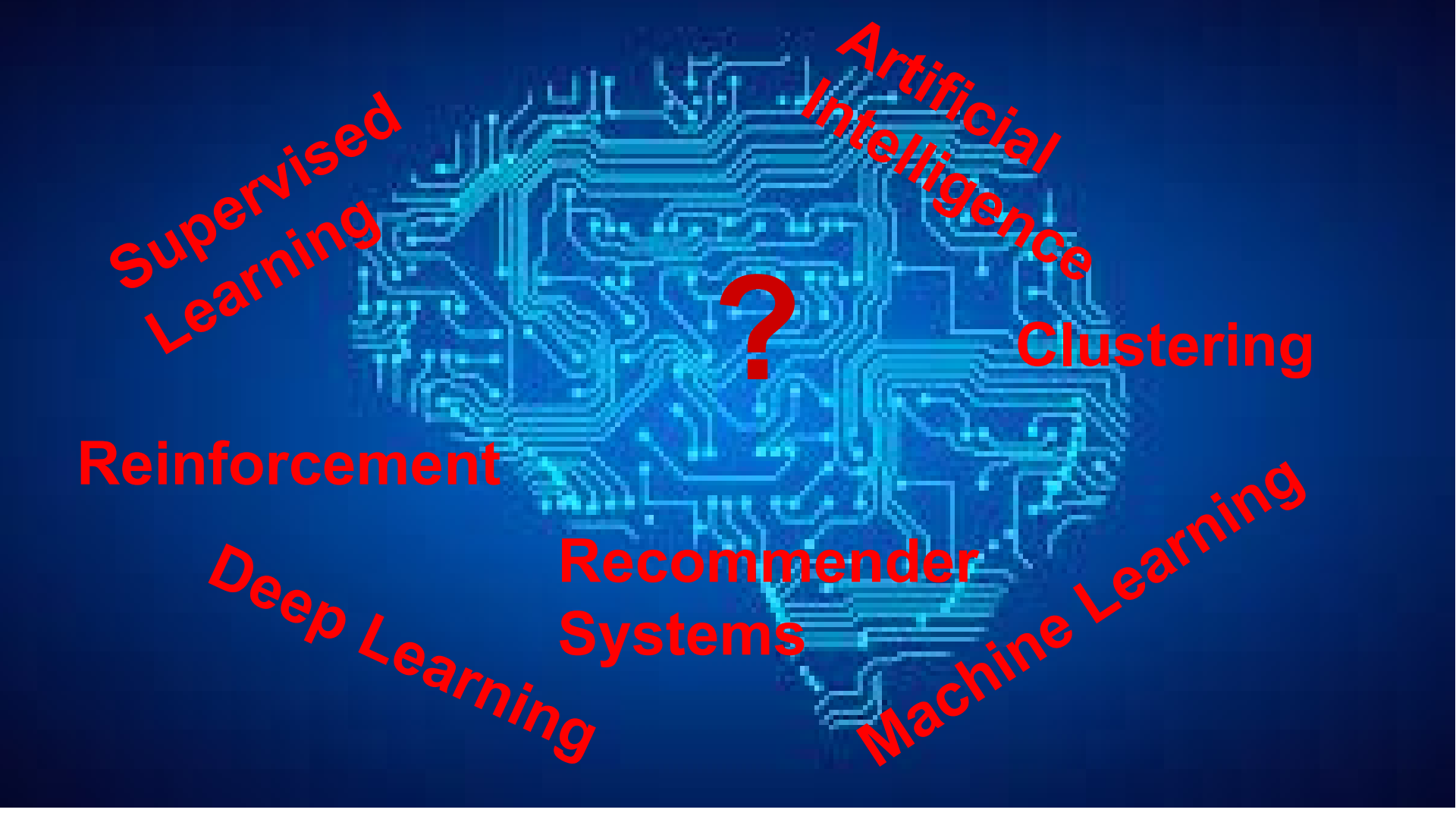
Machine Learning, AI, and All That

Sean Robinson


Supervised Learning
Artificial Intelligence
?
Clustering
Reinforcement
Recommender Systems
Deep Learning
Machine Learning

What is AI? Can machines learn like a human being? As importantly, should they be allowed to?

“I believe that at the end of the century the use of words and general educated opinion will have altered so much that one will be able to speak of machines thinking without expecting to be contradicted.”
— Alan Turing, Computing machinery and intelligence

“Human beings, viewed as behaving systems, are quite simple. The apparent complexity of our behavior over time is largely a reflection of the complexity of the environment in which we find ourselves.”
— Herbert A. Simon, The Sciences of the Artificial

“Whoever perceives that robots and artificial intelligence are merely here to serve humanity, think again. With virtual domestic assistants and driverless cars just the latest in a growing list of applications, it is we humans who risk becoming dumbed down and ultimately subservient to machines.”
— Alex Morritt, Impromptu Scribe

Modeling - the “classic science” way

Models

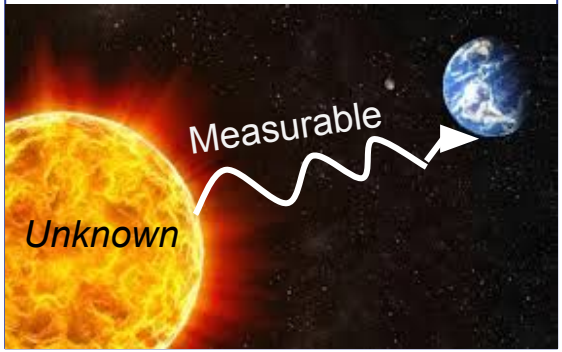
Describe the world in some specific way

Turn input parameters into estimated measurements

Many purposes

Usage

Say we want to measure *inside* the sun..



The Issue

Requires deep expertise

Can be super complex

Doesn’t catch unknown-unknowns

What happens when we have data with no model?

We need a better solution!

A human can learn from experience, why not a computer?

Many problems have data, but no model. Or known dynamics that are too complex or rapidly changing.

Learn Directly from data - but how?

- “We sure would like it if we had some nice general models to use....”
- Two things were happening:
  - Rapidly advancing computer speed
  - New ways of thinking about models and data analysis
- Sometimes there were problems that only had data, not any understood model, and they still needed answers:
  - What movies might you like?
  - How can we make a car drive itself?
  - Who's doing financial fraud?
- Lots of data possible, but the dynamics are poorly understood

A good approach: Machine Learning

Conceptually makes sense...

Let's find some methods that don't need expertise on the way in:

They could automate this kind of investigation

These approaches will start from the data, not the dynamics

Is really a whole set of approaches

Supervised machine learning:

- Learn from the past to predict the future

Unsupervised machine learning:

- Characterize data - compare, compress or cluster events

Each of which have multiple methods:

- Supervised Methods:
  - Regression (predict a value)
  - Classification (predict a class)
- Unsupervised Methods:
  - Clustering
  - Recommenders

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Implementation - Supervised Machine Learning

Let’s try to predict a class: Who will get an illness?

Variables

Individuals

	Gender (M/F)	Age	Weight (lbs.)	Height (in.)	Smoking (1=No, 2=Yes)	Race
Patient #1	M	59	175	69	1	White
Patient #2	F	67	140	62	2	Black
Patient #3	F	73	155	59	1	Asian
.	.	.	.	.	.	.
.	.	.	.	.	.	.
.	.	.	.	.	.	.
.	.	.	.	.	.	.
.	.	.	.	.	.	.
Patient #75	M	48	90	72	1	White

Let’s try to predict a class: Who will get an illness?

Variables

Individuals

	Gender (M/F)	Age	Weight (lbs.)	Height (in.)	Smoking (1=No, 2=Yes)	Race	Diabetes Test
Patient #1	M	59	175	69	1	White	Pos
Patient #2	F	67	140	62	2	Black	Pos
Patient #3	F	73	135	59	1	Asian	Neg
.	.	.	.	.	.	.	Neg
.	.	.	.	.	.	.	Pos
.	.	.	.	.	.	.	Neg
.	.	.	.	.	.	.	Neg
Patient #75	M	48	90	72	1	White	

Who is most likely to get an illness this year?

So how do we solve this problem in the real world?

- We want to take our training data and make some model that describes it – then use that to predict future outcomes.
- We could start with a model that just generalizes well, since a model is just a numerical construct used to try to simulate reality. Let's try a simple one:

Illness Risk

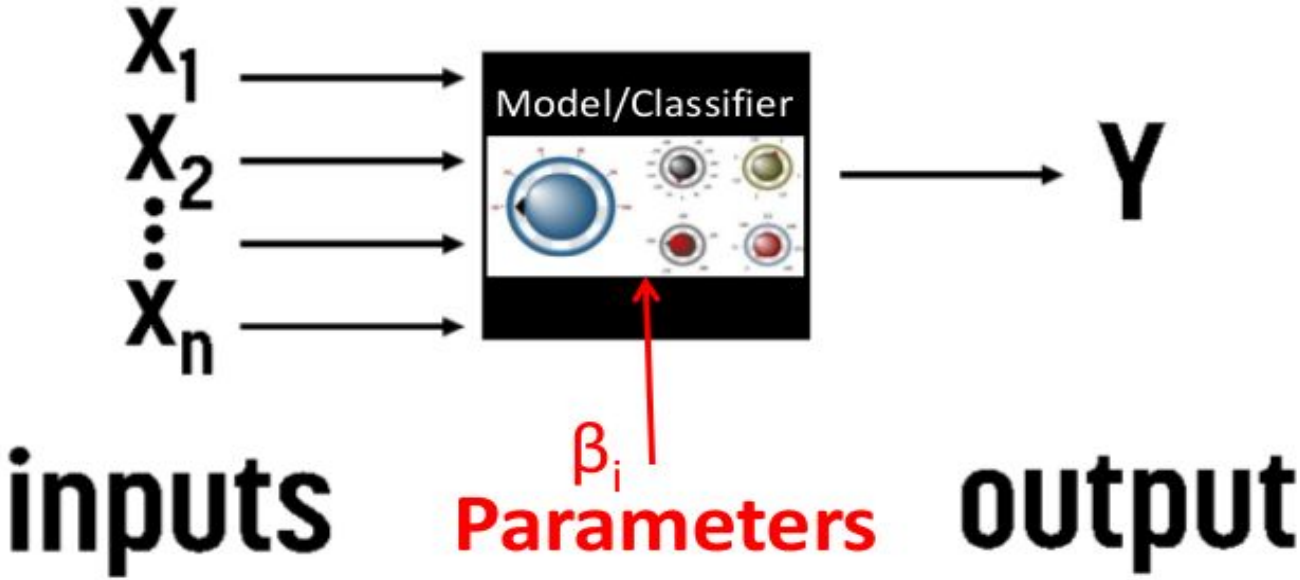
Measured Variables ("features")

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \ldots + \beta_n X_n + \varepsilon$$

- Why pick this one? It can represent a lot of physical situations.
- So let's see if it works! Wait though.....By even making up a model, I've added something...parameters!

Parameters

- The "knobs" you turn on the model.

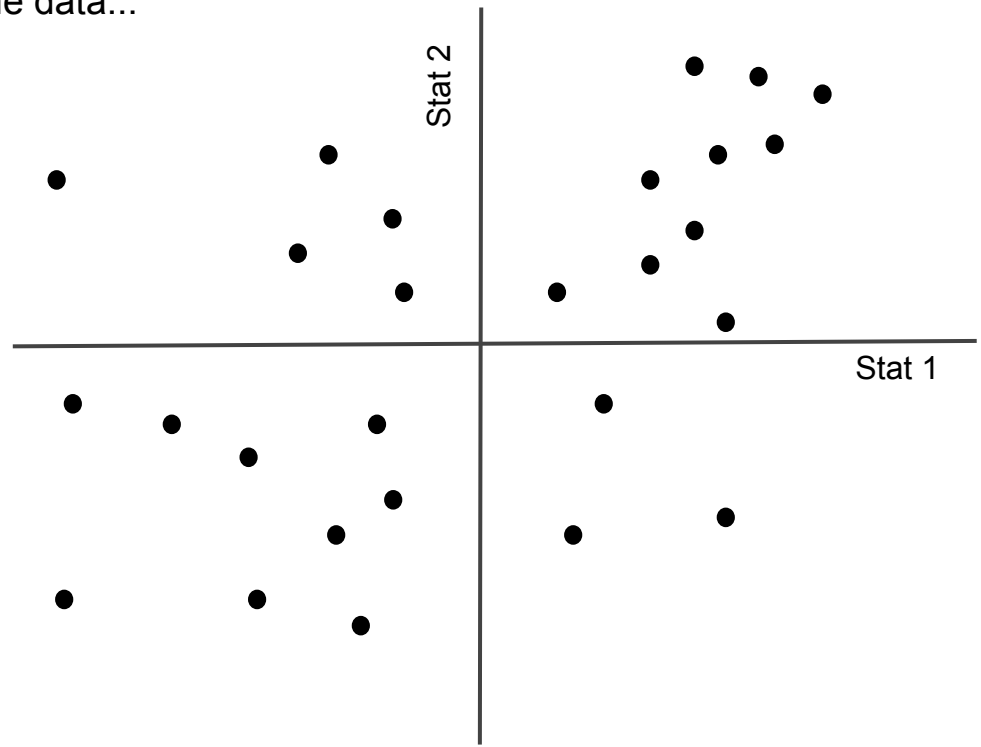


Putting it together:

- The Dark Art of Supervised Machine Learning is basically made of three actions:
  - Picking a model with some parameters that describe the data (Don't worry, there are lots to pick from that describe many possible situations).
  - Fitting the best parameters to optimize performance
  - Seeing how well our model really describes the world, so we can use it to predict.

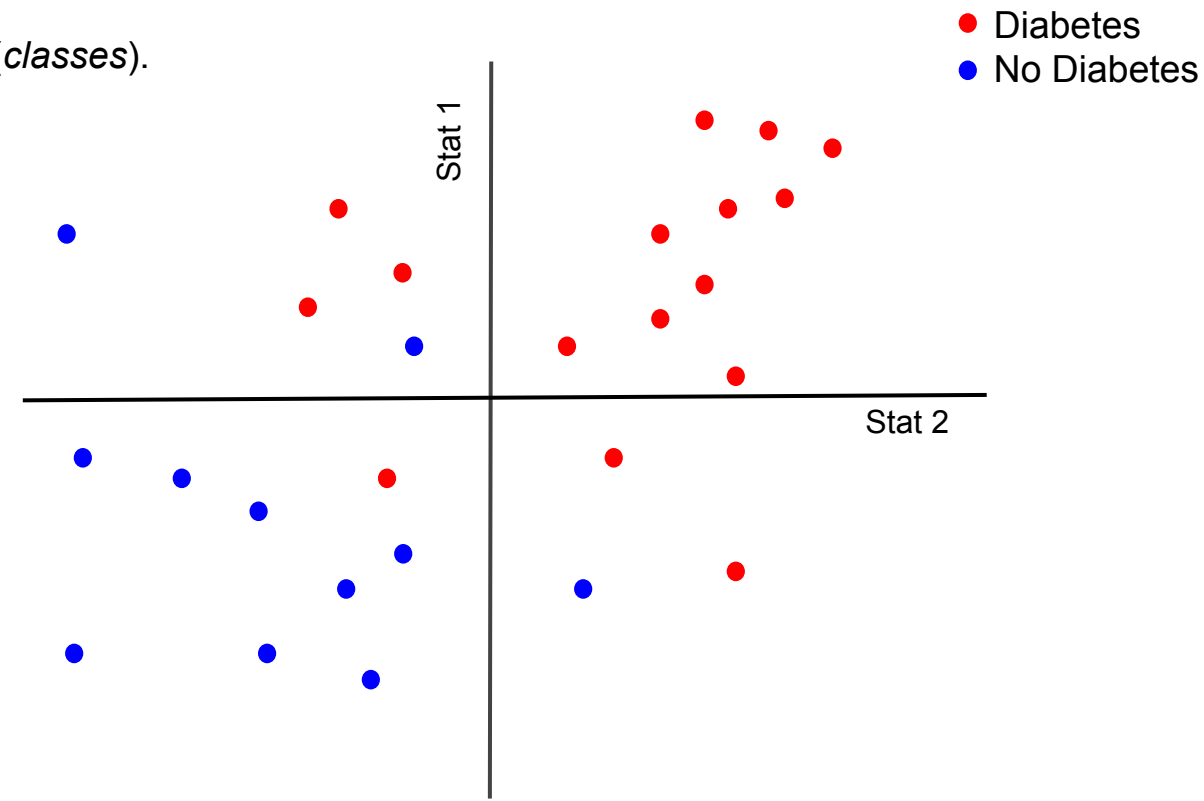
The beating heart of supervised machine learning:

Start by examining the data...



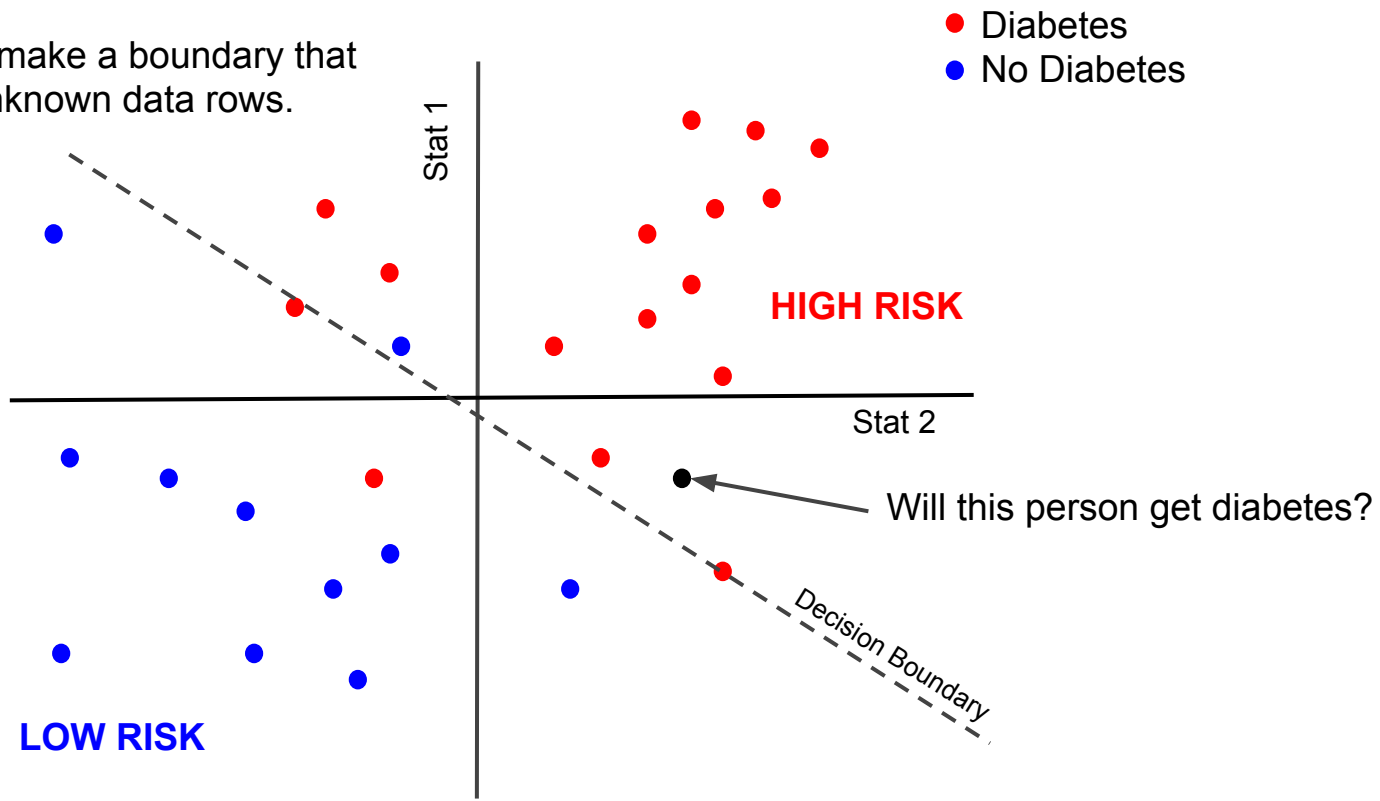
The beating heart of supervised machine learning:

Add the past labels (*classes*).



The beating heart of supervised machine learning:

To answer, we try to make a boundary that *classifies* new and unknown data rows.



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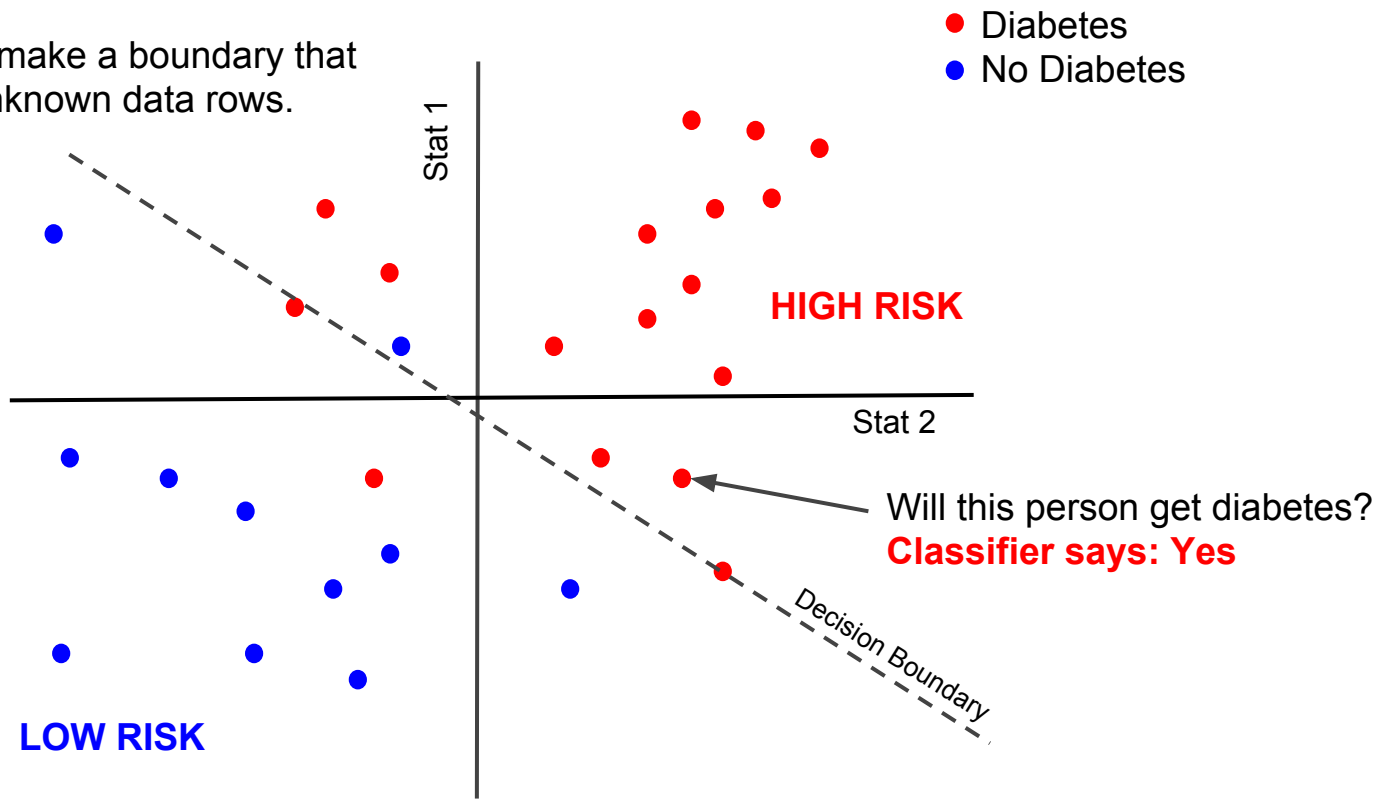
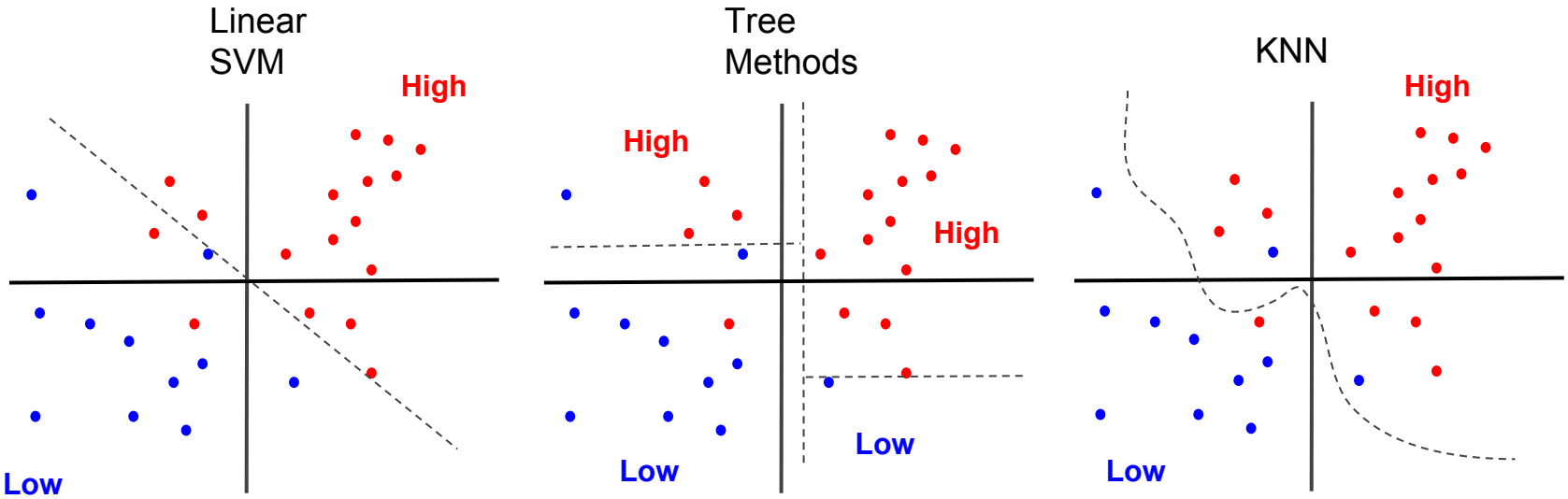


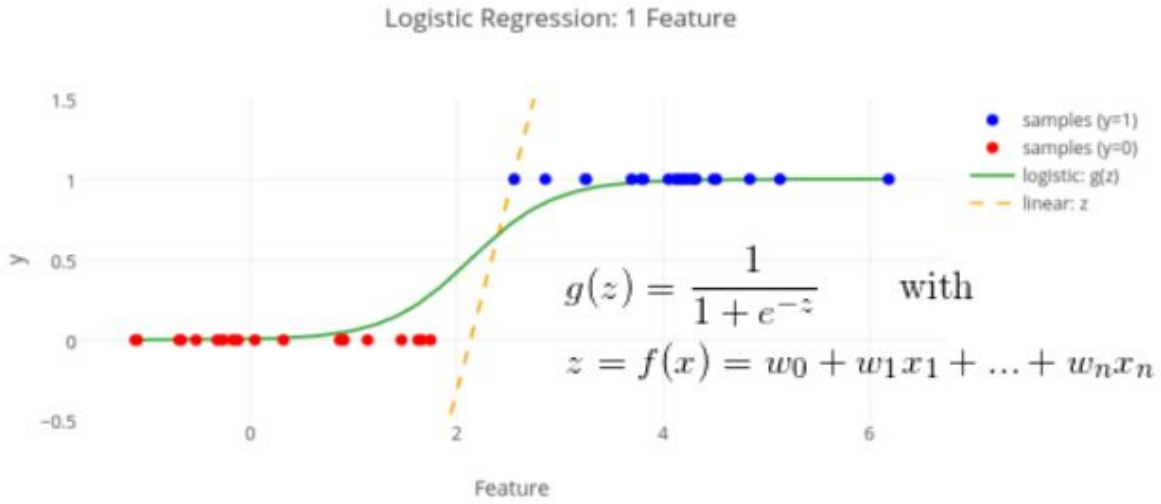
Decision boundaries come in many flavors:



Every model has parameters - these show up as things like slope or maximum number of spatial cuts!

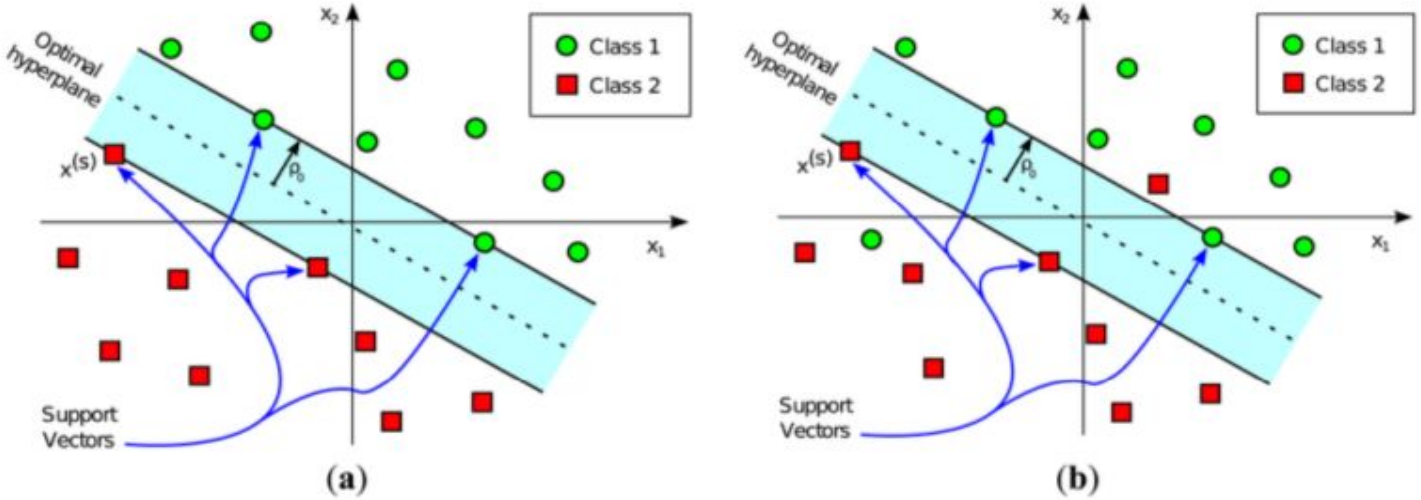
Logistic Regression

- The training data only has binary inputs, after all…here's a pre-made way to use that
- Robust, well understood, a great baseline



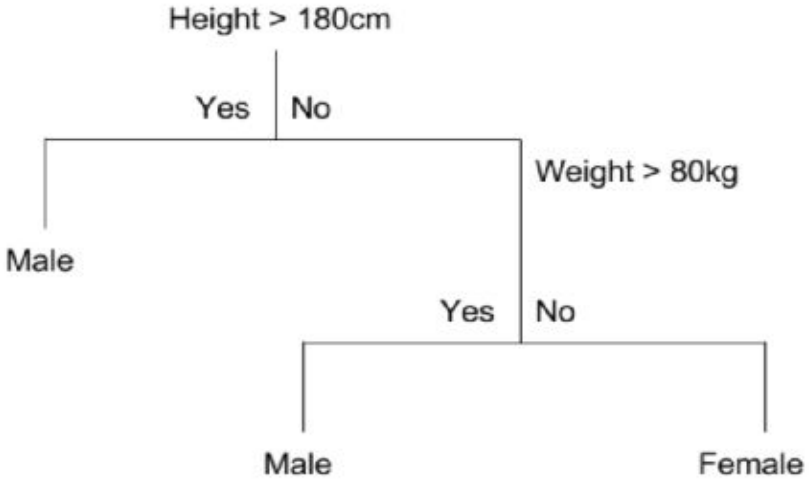
Support Vector Machines

- Attempt to optimally "cut" the variable space into binary classes
- Each row of data corresponds to one location, data are classified according to their distance from the fit boundary
- Can be made nonlinear to try to fit more complex boundaries



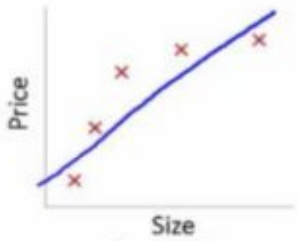
Tree Methods

- "Trees" are constructed using methods to choose decision points to best separate results
- Metrics applied to recursively separate set
- Many variants of this sort of method



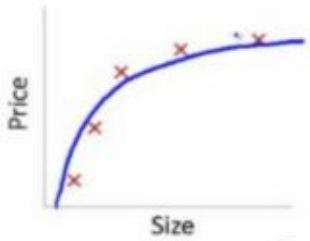
Oh S#!t, Overfit!

- Wait though, there's one more little problem – if we used all the data to train, how do we know how well it works?
- If I fit with a data set and then also check using it...



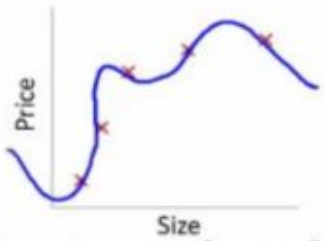
$\theta_0 + \theta_1 x$

High bias
(underfit)



$\theta_0 + \theta_1 x + \theta_2 x^2$

"Just right"

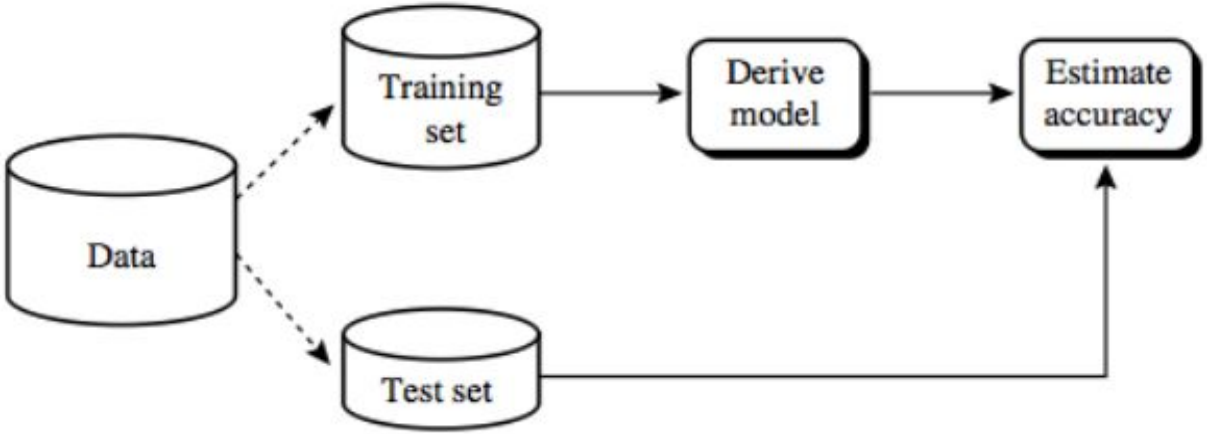


$\theta_0 + \theta_1 x + \theta_2 x^2 + \theta_3 x^3 + \theta_4 x^4$

High variance
(overfit)

What to do about overfit?

- The usual answer: cut the data up into some pieces and hold some back randomly. Our method had better work on data it wasn't explicitly trained on!



A Brief Review and applications:

A brief review

- Get your data and dep/ind variables
- Explore the data a little.
- Choose a model
- consider the parameter space
- pick a performance metric to optimize
- wander around the parameter space until you find the best performing spot
- use that "trained" model to predict future data.

So let’s put this in action!

- Enough talk - let’s make our own classifier and see how it works!
- We’ll use the built-in datasets and libraries in R for this purpose - other fine contenders include python and scikit-learn (sklearn)
- Spoilers: Most of the algorithms we’ll actually use are pre-canned and work pretty well. So the trick is in comparing them and knowing when they seem to be working!

A little prep first...

- Get R and Rstudio
- Install the following packages:
  - caret
  - class
  - gmodels
  - E1071
- Open a new file and put these in - we'll need them!
  - `library(class)`
  - `library(gmodels)`
  - `library(caret)`

The “Iris” data set

- measured in 1936 - four values measured from 3 populations of flower
- Data taken on a single day
- Often used to demonstrate ML applications
- Included with scikit-learn as well as R and can be found in other places

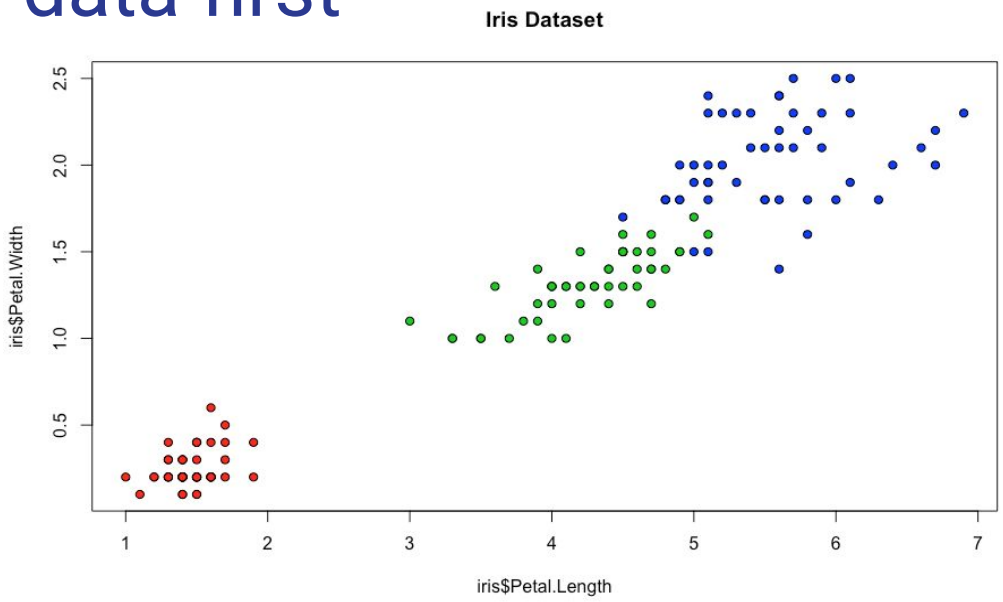
Credit where credit’s due...

In the grand tradition of pedagogy and engineering, we’re going to steal some code from a place it was done well in the past.

Take a look here for some of this approach spelled out in code:
https://www.datacamp.com/community/tutorials/machine-learning-in-r

Iris Versicolor | Iris Setosa | Iris Virginica

Let’s take a look at the data first



Enter the following:

```
iris
table(iris$Species)
```

Then have a look:

```
print('Examining the Iris Dataset')
plot(iris$Sepal.Length, iris$Sepal.Width, pch=21,
bg=c("red","green3","blue")[unclass(iris$Species)],
main="Iris Dataset")
```

Okay….but now try the petal width and length:
Consider other fun stuff like `print(summary(iris))`

R makes a lot of things really easy...

- Looks like there's a correlation between two variables, let's explore that:
  - `cor(iris$Petal.Length, iris$Petal.Width)`
- The whole data set has a high correlation. But what about just one of the species of flower?
  - `x=levels(iris$Species)`
  - `print(x[1])`
  - `print(cor(iris[iris$Species==x[1],1:4]))`
  - Hmm, maybe not so much. Try the other species?
- Some valuable outcomes here - it's unclear if the relationship holds within a species, but it looks like it does *between* species...

A quick aside: Normalization

- Normalization is the process of rescaling data so it falls in approximately the same range
- Why? Well, we're about to deploy KNN, and that has a "distance" in it…
- If one variable range is really broad, that will dominate the distance.
- Take a look at `print(summary(iris))`
  - Here the ranges are similar and it isn't really necessary, but keep it in mind!

```
 Sepal.Length    Sepal.Width     Petal.Length    Petal.Width          Species  
Min.   :4.300   Min.   :2.000   Min.   :1.000   Min.   :0.100   setosa    :50  
1st Qu.:5.100   1st Qu.:2.800   1st Qu.:1.600   1st Qu.:0.300   versicolor:50  
Median :5.800   Median :3.000   Median :4.350   Median :1.300   virginica :50  
Mean   :5.843   Mean   :3.057   Mean   :3.758   Mean   :1.199                  
3rd Qu.:6.400   3rd Qu.:3.300   3rd Qu.:5.100   3rd Qu.:1.800                  
Max.   :7.900   Max.   :4.400   Max.   :6.900   Max.   :2.500                  
```

Remember to partition the data!

- Split into two parts, test and training, to allow for testing performance afterward
  - `set.seed(1234)`
  - `ind <- sample(2, nrow(iris), replace=TRUE, prob=c(0.67, 0.33))`
  - `iris.training <- iris[ind==1, 1:4]`
  - `print(head(iris.training))`
  - `iris.test <- iris[ind==2, 1:4]`
  - `print(head(iris.test))`
- Hey wait, we only got the columns 1-4. Why?
  - `iris.trainLabels <- iris[ind==1,5]`
  - `iris.testLabels <- iris[ind==2, 5]`

And now the moment of truth…..

- To begin the arduous work of making a KNN classifier, we…
  - `iris_pred <- knn(train = iris.training, test = iris.test, cl = iris.trainLabels, k=3)`
  - `print(iris_pred)`
- Wait, that's it? We've guessed the classes of some 'new' flowers?
- Well, almost.

```
[1] "Examining the Iris Dataset"
[1] "prediction!"
 [1] setosa     setosa     setosa     setosa     setosa     setosa     setosa     setosa     setosa     setosa     setosa     setosa     versicolor versicolor
[15] versicolor versicolor versicolor versicolor versicolor versicolor versicolor versicolor versicolor versicolor virginica  virginica  virginica  virginica
[29] versicolor virginica  virginica  virginica  virginica  virginica  virginica  virginica  virginica  virginica  virginica  virginica
Levels: setosa versicolor virginica
```

Let’s take a look at performance

- If we put the actual and predicted test data side by side….
  - `irisTestLabels <- data.frame(iris.testLabels)`
  - `merge <- data.frame(iris_pred, iris.testLabels)`
  - `names(merge) <- c("Predicted Species", "Observed Species")`
  - `print(head(merge,50))`
- How about something more quantitative?
  - `abc = confusionMatrix(iris.testLabels,iris_pred)`
  - `print(abc)`

There’s so much more!!

- The caret pakage is also excellent and allows for the same kind of thing:
  - `model_knn <- train(iris.training[, 1:4], iris.training[, 5], method='knn', preProcess=c("center", "scale"))`
- `Easy to train, keep the model, and use it to predict later!`

- But lots of other methods can go here, not just KNN!
  - SVM
  - Random Forest
  - Logistic Regression
  - And others!

Some take-home points:

Machine Learning is a big topic!

We just talked about a piece - but that piece is often very effective!

- Overfitting
- Goodness of fit
- Data partitioning
- Gradient Ascent and fit
- Generative models
- Adversarial Learning

And many more things are still out there...

Homework

Consider data analysis needs that follow these patterns

Consider one of the many good online courses!

Some Useful References

- http://scikit-learn.org/stable/tutorial/basic/tutorial.html
- http://stackoverflow.com/questions/9480605/what-is-the-relation-between-the-number-of-support-vectors-and-training-data-and
- http://www.turingfinance.com/perils-optimization-in-investment-management/
- https://sites.google.com/a/kingofat.com/wiki/data-mining/classification
- https://www.quora.com/What-are-the-advantages-of-different-classification-algorithms
- https://florianhartl.com/logistic-regression-geometric-intuition.html
- http://nlp.stanford.edu/IR-book/html/htmledition/nonlinear-svms-1.html
- http://www.saedsayad.com/naive_bayesian.htm
- http://www.r2d3.us/visual-intro-to-machine-learning-part-1/
- http://scikit-learn.org/stable/modules/tree.html

Questions?